# 314707003_15.20.R

## CDH

## 2026-05-19

```
rm(list=ls())
#install.packages("plm")
library(POE5Rdata)
```

```
##
## 載入套件：'POE5Rdata'
```

```
## 下列物件被遮斷自 'package:datasets':
##
##     euro
```

```
library(plm)
```

```
## Warning: 套件 'plm' 是用 R 版本 4.5.3 來建造的
```

```
library(car)
```

```
## Warning: 套件 'car' 是用 R 版本 4.5.3 來建造的
```

```
## 載入需要的套件：carData
```

```
## Warning: 套件 'carData' 是用 R 版本 4.5.3 來建造的
```

```
library(lmtest)
```

```
## Warning: 套件 'lmtest' 是用 R 版本 4.5.3 來建造的
```

```
## 載入需要的套件：zoo
```

```
## Warning: 套件 'zoo' 是用 R 版本 4.5.3 來建造的
```

```
##
## 載入套件：'zoo'
```

```
## 下列物件被遮斷自 'package:base':
##
##     as.Date, as.Date.numeric
```

```
library(sandwich)
```

```
## Warning: 套件 'sandwich' 是用 R 版本 4.5.3 來建造的
```

```
data(star)
#str(star)
#15.20(a)
ols_mod <- lm(
  readscore ~ small + aide + tchexper +
    boy + white_asian + freelunch,
  data = star
)

coeftest(
  ols_mod,
  vcov = vcovHC(ols_mod, type = "HC1")
)
```

```
##
## t test of coefficients:
##
##               Estimate Std. Error  t value  Pr(>|t|)
## (Intercept) 437.76425    1.31238 333.5646 < 2.2e-16 ***
## small         5.82282    0.99601   5.8461 5.308e-09 ***
## aide          0.81784    0.93280   0.8768    0.3807
## tchexper      0.49247    0.07128   6.9090 5.409e-12 ***
## boy          -6.15642    0.79656  -7.7287 1.273e-14 ***
## white_asian   3.90581    0.89098   4.3837 1.187e-05 ***
## freelunch   -14.77134    0.85637 -17.2488 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
#15.20(b)
fe_mod <- lm(
  readscore ~ small + aide + tchexper +
    boy + white_asian + freelunch +
    factor(schid),
  data = star
)

coeftest(
  fe_mod,
  vcov = vcovHC(fe_mod, type = "HC1")
)[1:7, ]
```

```
##                 Estimate Std. Error    t value     Pr(>|t|)
## (Intercept) 411.1626296 2.37842242 172.871996 0.000000e+00
## small         6.4902305 0.93102600   6.971052 3.503010e-12
```

```
## aide           0.9960875 0.86127432   1.156528 2.475141e-01
## tchexper       0.2855668 0.07114639   4.013792 6.052000e-05
## boy           -5.4559412 0.73104341  -7.463225 9.716472e-14
## white_asian    8.0280192 1.53786343   5.220242 1.850422e-07
## freelunch    -14.5935724 0.84646565 -17.240596 5.786271e-65
```

```
#15.20(c)
linearHypothesis(
  fe_mod,
  grep("factor\\(schid\\)", names(coef(fe_mod)), value = TRUE),
  vcov = vcovHC(fe_mod, type = "HC1")
)
```

```
##
## Linear hypothesis test:
## factor(schid)123056 = 0
## factor(schid)128068 = 0
## factor(schid)128076 = 0
## factor(schid)128079 = 0
## factor(schid)130085 = 0
## factor(schid)159171 = 0
## factor(schid)161176 = 0
## factor(schid)161183 = 0
## factor(schid)162184 = 0
## factor(schid)164198 = 0
## factor(schid)165199 = 0
## factor(schid)166203 = 0
## factor(schid)168211 = 0
## factor(schid)168214 = 0
## factor(schid)169219 = 0
## factor(schid)169229 = 0
## factor(schid)169231 = 0
## factor(schid)169280 = 0
## factor(schid)170295 = 0
## factor(schid)173312 = 0
## factor(schid)176329 = 0
## factor(schid)180344 = 0
## factor(schid)189378 = 0
## factor(schid)189382 = 0
## factor(schid)189396 = 0
## factor(schid)191411 = 0
## factor(schid)193422 = 0
## factor(schid)193423 = 0
## factor(schid)201449 = 0
## factor(schid)203452 = 0
## factor(schid)203457 = 0
## factor(schid)205488 = 0
## factor(schid)205489 = 0
## factor(schid)205490 = 0
## factor(schid)205491 = 0
## factor(schid)205492 = 0
## factor(schid)208501 = 0
## factor(schid)208503 = 0
## factor(schid)209510 = 0
## factor(schid)212522 = 0
## factor(schid)215533 = 0
## factor(schid)216536 = 0
```

```
## factor(schid)218562 = 0
## factor(schid)221571 = 0
## factor(schid)221574 = 0
## factor(schid)225585 = 0
## factor(schid)228606 = 0
## factor(schid)230612 = 0
## factor(schid)231616 = 0
## factor(schid)234628 = 0
## factor(schid)244697 = 0
## factor(schid)244708 = 0
## factor(schid)244723 = 0
## factor(schid)244727 = 0
## factor(schid)244728 = 0
## factor(schid)244736 = 0
## factor(schid)244745 = 0
## factor(schid)244746 = 0
## factor(schid)244755 = 0
## factor(schid)244764 = 0
## factor(schid)244774 = 0
## factor(schid)244776 = 0
## factor(schid)244780 = 0
## factor(schid)244796 = 0
## factor(schid)244799 = 0
## factor(schid)244801 = 0
## factor(schid)244806 = 0
## factor(schid)244818 = 0
## factor(schid)244831 = 0
## factor(schid)244839 = 0
## factor(schid)252885 = 0
## factor(schid)253888 = 0
## factor(schid)257899 = 0
## factor(schid)257905 = 0
## factor(schid)259915 = 0
## factor(schid)261927 = 0
## factor(schid)262937 = 0
## factor(schid)264945 = 0
##
## Model 1: restricted model
## Model 2: readscore ~ small + aide + tchexper + boy + white_asian + freelun
ch +
##     factor(schid)
##
## Note: Coefficient covariance matrix supplied.
##
##   Res.Df Df      F    Pr(>F)
## 1   5759
## 2   5681 78 20.334 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
#15.20(d)
star$obs <- ave(star$readscore, star$schid, FUN = seq_along)

star_p <- pdata.frame(
  star,
  index = c("schid", "obs")
)
```

```
pool_mod <- plm(
  readscore ~ small + aide + tchexper +
    boy + white_asian + freelunch,
  data = star_p,
  model = "pooling"
)

re_mod <- plm(
  readscore ~ small + aide + tchexper +
    boy + white_asian + freelunch,
  data = star_p,
  model = "random"
)

coeftest(
  re_mod,
  vcov = vcovHC(re_mod, type = "HC1")
)
```

```
##
## t test of coefficients:
##
##               Estimate Std. Error  t value  Pr(>|t|)
## (Intercept) 436.12677    2.04477 213.2890 < 2.2e-16 ***
## small         6.45872    1.63589   3.9481 7.970e-05 ***
## aide          0.99215    1.38238   0.7177   0.47297
## tchexper      0.30268    0.13570   2.2305   0.02575 *
## boy          -5.51208    0.70447  -7.8244 6.028e-15 ***
## white_asian   7.35048    1.74429   4.2140 2.547e-05 ***
## freelunch   -14.58433    1.05731 -13.7939 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
# LM test for random effects
plmtest(pool_mod, type = "bp")
```

```
##
##  Lagrange Multiplier Test - (Breusch-Pagan)
##
## data:  readscore ~ small + aide + tchexper + boy + white_asian + freelunch
## chisq = 6677.4, df = 1, p-value < 2.2e-16
## alternative hypothesis: significant effects
```

```
#15.20(e)
fe_plm <- plm(
  readscore ~ small + aide + tchexper +
    boy + white_asian + freelunch,
  data = star_p,
  model = "within"
)

phtest(fe_plm, re_mod)
```

```
##
##  Hausman Test
##
## data:  readscore ~ small + aide + tchexper + boy + white_asian + freelunch
## chisq = 13.809, df = 6, p-value = 0.03184
## alternative hypothesis: one model is inconsistent
```

```
# 如果只看 BOY 的 FE vs RE 差異
coef(fe_plm)["boy"]
```

```
##       boy
## -5.455941
```

```
coef(re_mod)["boy"]
```

```
##       boy
## -5.512081
```

```
#15.20(f)
star$m_small <- ave(star$small, star$schid, FUN = mean)
star$m_aide <- ave(star$aide, star$schid, FUN = mean)
star$m_tchexper <- ave(star$tchexper, star$schid, FUN = mean)
star$m_boy <- ave(star$boy, star$schid, FUN = mean)
star$m_white_asian <- ave(star$white_asian, star$schid, FUN = mean)
star$m_freelunch <- ave(star$freelunch, star$schid, FUN = mean)

# Mundlak model
mundlak_mod <- lm(
  readscore ~ small + aide + tchexper +
    boy + white_asian + freelunch +
    m_small + m_aide + m_tchexper +
    m_boy + m_white_asian + m_freelunch,
  data = star
)

# 看係數
coeftest(
  mundlak_mod,
  vcov = vcovCL(mundlak_mod, cluster = ~ schid)
)
```

```
##
## t test of coefficients:
##
##                Estimate Std. Error  t value  Pr(>|t|)
## (Intercept)   464.07945   20.10862  23.0786 < 2.2e-16 ***
## small           6.56177    1.66424   3.9428 8.150e-05 ***
## aide            1.09202    1.40132   0.7793   0.43585
## tchexper        0.29460    0.13896   2.1200   0.03405 *
## boy            -5.40859    0.71842  -7.5284 5.938e-14 ***
## white_asian     8.19637    1.88181   4.3556 1.350e-05 ***
## freelunch     -14.64249    1.08372 -13.5113 < 2.2e-16 ***
```

```
## m_small       -22.55156   20.91198  -1.0784   0.28090
## m_aide         10.88891   19.59374   0.5557   0.57841
## m_tchexper      0.82536    0.57958   1.4241   0.15448
## m_boy         -51.23797   23.81581  -2.1514   0.03148 *
## m_white_asian  -7.02301    5.35647  -1.3111   0.18987
## m_freelunch    -3.62902    7.18736  -0.5049   0.61364
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
# Mundlak test:
# H0: school averages jointly insignificant
linearHypothesis(
  mundlak_mod,
  c(
    "m_small = 0",
    "m_aide = 0",
    "m_tchexper = 0",
    "m_boy = 0",
    "m_white_asian = 0",
    "m_freelunch = 0"
  ),
  vcov = vcovCL(mundlak_mod, cluster = ~ schid)
)
```

```
##
## Linear hypothesis test:
## m_small = 0
## m_aide = 0
## m_tchexper = 0
## m_boy = 0
## m_white_asian = 0
## m_freelunch = 0
##
## Model 1: restricted model
## Model 2: readscore ~ small + aide + tchexper + boy + white_asian + freelun
ch +
##     m_small + m_aide + m_tchexper + m_boy + m_white_asian + m_freelunch
##
## Note: Coefficient covariance matrix supplied.
##
##   Res.Df Df      F Pr(>F)
## 1   5695
## 2   5689  6 2.2537 0.0356 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```